UBT844: ENVIRONMENTAL BIOTECHNOLOGY

L	T	P	Cr
3	0	0	3.0

Course Objective: The course content aims to make the students understand how biotechnology can help in monitoring or removing the pollutants and developing an understanding of new trends such as biofuels, renewable energy sources, or microbial technologies which can minimize the harmful impact of pollutants in the environment.

Syllabus

Introduction to Environmental Pollutants and Scope of Environmental Biotechnology: Water, soil and waste water their sources and effects. Application of biotechnology in environment protection

Biological Wastewater Treatment: Principles and Microbiology of wastewater treatment, unit operations: Aerobic process (Activated sludge, Oxidation ditches, Trickling filters, rotating discs, rotating drums, oxidation ponds). Anaerobic processes (Anaerobic filters, Up-flow anaerobic sludge blanket reactors), and other emerging biotechnological processes in waste water treatment for municipal, industrial wastewater

Solid Waste Management: Landfills, recycling and processing of organic residues, composting technologies, Biofuel production: Biogas, bioethanol, biohydrogen and biodiesel

Bioremediation and Biodegradation: Introduction and types of bioremediations, Microbial systems for heavy metal accumulation, Biosorption & detoxification mechanisms, Metal bioleaching and bio-oxidation In-situ and Ex-situ technologies, effect of chemical structure on biodegradation, recalcitrance, co-metabolism and biotransformation. Factors affecting biodegradation, microbial degradation of xenobiotic compounds and hydrocarbons: long chain aliphatic, aromatic, halogenated, sulfonated compounds, surfactants, pesticides and oil spills.

Environmental Genetics: Plasmid borne metabolic activities, bioaugmentation, release of genetically engineered organisms in environment, Biosensor technology for monitoring pollutants.

Course Learning Objectives (CLO)
The students will be able to:

1. Comprehend environmental issues and role of biotechnology in the clean-up of contaminated environments.

2. Comprehend fundamentals of biodegradation, biotransformation and bioremediation of organic contaminants and toxic metals.
3. Apply biotechnological processes in wastewater and solid waste management.
4. Comprehend biofuels/bioenergy systems; attributes for biofuel / bioenergy production.
5. Demonstrate innovative biotechnological interventions to combat environmental challenges.

Text Books

1. Rittmann, B. and McCarty, P., Environmental Biotechnology: Principles and Applications, McGraw-Hill (2006)
2. Environmental Biotechnology, B.C. Bhattacharya & Ritu Banerjee, Oxford Press, 2007.

Reference Books

1. Scargg, A., Environmental Biotechnology, Longman (1999).
2. Wainwright, M., An Introduction to Environmental Biotechnology, Kluwer Academic Press (1999).
3. Environmental Microbiology & Biotechnology, D.P. Singh, S.K. Dwivedi, New Age International Publishers, 2004.

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	25-35
2	EST	35-45
3	Sessional (May include Assignments//Quizzes)	30